Local Existence and Covariance Lemma

Detailed Sketch for gPCT Reconstruction

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Statement (target lemma)

Let $s(\tau)$ be a smooth slope history on a compact interval such that

$$\left|\int s\, d\tau\right| \le \varepsilon \qquad \text{and} \qquad |\mathcal{H}(s) + 1/(2\pi)| \le \varepsilon.$$

Then there exists a Lorentzian metric $g_{\mu\nu}$, defined in a neighbourhood of the worldline and unique up to isometry, such that

1. $\frac{1}{2}\ln(-g_{00})$ reproduces s along the worldline,
2. $|G_{\mu\nu}(g)| \le C\varepsilon$ for a dimensionless constant C independent of any absolute scale,
3. the same bound holds in every gauge.

Detailed Sketch

Setup

Let $I \subset \mathbb{R}$ be a compact interval and $s \in C^{2,\alpha}(I)$ satisfy the residual bounds above. Write $\phi_g(\tau) = \int^\tau s$. We work throughout with dimensionless quantities.

A. Gaussian-chart existence (contraction mapping)

Introduce Gaussian normal coordinates (τ, x^i) in a tubular neighbourhood of the worldline so that the worldline is $x^i = 0$ and $g_{0i} = 0$. The condition that the recursion scalar remain within ε of its attractor off the worldline implies

$$\partial_i \phi_g = \mathcal{O}(\varepsilon), \qquad \partial_i \partial_j \phi_g = \mathcal{O}(\varepsilon).$$

Let $B_r \subset \mathbb{R}^3$ be the coordinate ball of radius r transverse to the worldline. Define the Banach spaces

$$X = C^{2,\alpha}(B_r), \qquad Y = C^{0,\alpha}(B_r)$$

of symmetric 2-tensors, and the map

$$F : X \times [0, \varepsilon_0) \to Y,$$

$$F(\gamma, \varepsilon) = \mathrm{Ric}(\gamma) - Q(\partial\phi_g, \partial^2\phi_g) - \varepsilon\, R(\gamma, \phi_g),$$

where Q is quadratic and R collects the remaining lower-order terms linear in the residual slope.

The linearisation of F at the flat metric δ is the Lichnerowicz operator $L = \Delta_L$. Standard Schauder theory supplies a constant $C_S = C_S(\alpha)$ such that

$$\|L^{-1} f\|_{C^{2,\alpha}} \le C_S \|f\|_{C^{0,\alpha}}.$$

Choose $r_0 > 0$ and $\varepsilon_0 > 0$ small enough that the quadratic remainder satisfies

$$\|Q(\gamma_1) - Q(\gamma_2)\|_{C^{0,\alpha}} \le \frac{1}{2C_S} \|\gamma_1 - \gamma_2\|_{C^{2,\alpha}}$$

whenever $\|\gamma_k - \delta\|_{C^{2,\alpha}} \le 1$. Then the map

$$T(\gamma) = \delta - L^{-1}\big(Q(\gamma) + \varepsilon R(\gamma, \phi_g)\big)$$

is a contraction on that ball. The Banach fixed-point theorem yields a unique fixed point $\gamma_\varepsilon \in X$ obeying

$$\|\gamma_\varepsilon - \delta\|_{C^{2,\alpha}} \le C\varepsilon.$$

The metric

$$g = -e^{2\phi_g}\, d\tau^2 + \gamma_\varepsilon$$

therefore satisfies $|G_{\mu\nu}(g)| \le C'\varepsilon$ in the Gaussian chart, with C' dimensionless and independent of any absolute scale.

B. Unique continuation / covariance (Carleman estimate)

Let g and g' be two such metrics, possibly in different gauges, both realising the same ϕ_g on the worldline and both satisfying $|G_{\mu\nu}| \le C\varepsilon$. Their difference δg satisfies the linearised Einstein equation

$$P(\delta g) := (D\,\mathrm{Ein})_g(\delta g) = \mathcal{O}(\varepsilon).$$

Work in the Gaussian chart and introduce the Carleman weight $e^{\tau\phi}$ with $\phi(x) = |x|^2$ (strictly convex and vanishing on the worldline). For the linearised operator in harmonic gauge and for all sufficiently large τ,

$$\tau^3 \int_{B_r} |u|^2 e^{2\tau\phi}\, dx \le C \int_{B_r} |Pu|^2 e^{2\tau\phi}\, dx + C \int_{\partial B_r} \cdots .$$

Inserting the inhomogeneous equation, the bulk error of size $\mathcal{O}(\varepsilon)$ is absorbed by the left-hand side once τ exceeds a constant that depends only on the C^1 norm of the background metric. The resulting estimate yields

$$\|\delta g\|_{L^2(B_{r/2})} \le C\varepsilon.$$

Elliptic bootstrapping upgrades the bound to $C^{2,\alpha}$. Consequently, if δg and $\nabla\delta g$ vanish on the worldline, then $\delta g \equiv 0$ in the connected neighbourhood up to $\mathcal{O}(\varepsilon)$. The two metrics therefore determine isometric geometries, and the curvature bound is independent of the chart.

C. Continuous dependence

The solution map $\varepsilon \mapsto g_\varepsilon$ is continuous from $\mathbb{R}$ into $C^{2,\alpha}$. Hence $g_\varepsilon \to g_0$ as $\varepsilon \to 0$, where g_0 is an exact vacuum solution realising the original phase.

This completes the expanded sketch of the local-existence and covariance argument.

Status

The contraction-mapping formulation and the Carleman outline reduce local existence, the scale-free curvature estimate, and covariance to quantitative inequalities (Schauder constant, contraction threshold, Carleman absorption constant). These inequalities follow standard lines in the local theory of the Einstein equation, but the numerical factors and boundary terms have not been verified line-by-line. Until that verification is supplied, the lemma remains a detailed analytic outline rather than a fully checked theorem.